



## ✓ EXPERIMENT NO. 1

### BRAKE TEST ON SQUIRREL CAGE INDUCTION MOTOR

#### AIM

Conduct the brake test on 3 phase squirrel cage induction motor and plot the following performance characteristics.

- Electrical characteristics - Speed, line current, torque, power factor, efficiency & % slip Vs output power
  - Mechanical characteristics - Speed Vs Torque
- Also find the additional kVAR required to improve the power factor to 0.95 at various loads.

#### APPARATUS

| S. No | Name of the apparatus | Type | Range | Quantity |
|-------|-----------------------|------|-------|----------|
| 1     | Ammeter               |      |       |          |
| 2     | Voltmeter             |      |       |          |
| 3     | Wattmeter             |      |       |          |
| 4     | Tachometer            |      |       |          |
| 5     | TPDT                  |      |       |          |
| 6     | TPST                  |      |       |          |

#### PRINCIPLE

The two types of 3-phase induction motors are i) squirrel cage induction motor and ii) slip-ring induction motor. Three-phase squirrel cage induction motor is generally preferred because it is rugged in construction, requires less maintenance and is economical as compared to 3-phase slip ring induction motor. When the stator winding is connected to three phase ac supply, a rotating magnetic field is established in the air gap which rotates at synchronous speed.

Initially, rotor is stationary. Due to relative speed between the rotating magnetic field and stationary rotor conductors, an emf is induced in the rotor. As the rotor circuit is closed, currents will circulate through them. According to Lenz's Law, these induced currents will flow in such a direction so as to oppose the cause producing it. Here the cause is relative speed. In order to reduce the relative speed, the currents in the rotor produce a torque tending to rotate the rotor in the same direction of rotating field. At synchronous speed of the rotor, the relative speed is zero, no emf and no torque is developed, rotor tends to stop, hence rotor cannot attain synchronous speed. Motor runs at a speed slightly less than synchronous speed.